MATH-GA.2020: Numerical Methods II
Fourier Stability Analysis, Boundary Layers

Chuanwan (Christine) Wu (email:cw4459@nyu.edu)

May 14, 2024

1 Advection-Diffusion Process

Can be skipped, for later section explanations only.

1.1 PDE Derivation

Consider a density distribution $\vec{\rho}(\vec{x},t)$, put it varying with time:

$$\frac{d}{dt}\int \vec{\rho}(\vec{x},t)d\vec{x} + \int \vec{F}\cdot\vec{n}d\vec{l} = 0 \tag{1}$$

Define a domain Ω, and the boundary term $\partial\Omega$ with the conservation law holds

$$\int_\Omega \text{div}(\vec{F})d\vec{x} = \int_{\partial\Omega}\vec{F}\cdot\vec{n}d\vec{l} \Longrightarrow \int_\Omega\left(\frac{\partial\rho}{\partial t} + \text{div}(\vec{F})\right)d\vec{x} = 0 \tag{2}$$

Now consider a advection ($\vec{F_1}$) and a diffusion ($\vec{F_2}$) process happening in the domain Ω, we have:

$$\vec{F_1}(\vec{x},t) = \vec{v}\vec{\rho} = \vec{v}(\vec{x})\cdot\rho(\vec{x},t), \quad \vec{F_2}(\vec{x},t) = -\vec{\mu}(\vec{x})\cdot\vec{\nabla}\rho(\vec{x},t) \tag{3}$$

where both $\vec{v}(\vec{x})$ and $\vec{\mu}(\vec{x})$ are the drivers, but with the assumption that they don't change with time t. With $F = F_1 + F_2$, we get

$$\vec{F}(\vec{x},t) = \vec{F_1}(\vec{x},t) + \vec{F_2}(\vec{x},t) = \vec{v}(\vec{x})\cdot\rho(\vec{x},t) - \vec{\mu}(\vec{x})\cdot\vec{\nabla}\rho(\vec{x},t) \tag{4}$$

Since, we are in $x$domain only, we have $\vec{\nabla} = \partial_x$. Now, we want to see how the system change with $\text{div}(\vec{F})$, where $\text{div}(\vec{F}) = \partial x\vec{F} + \partial_t\vec{F}$ (t is also a dimension)

$$\begin{aligned}
\text{div}(\vec{F}) &= \partial_x\vec{F} + \partial_t\vec{F} \\
&= \partial_x(\vec{v}(\vec{x})\cdot\rho(\vec{x},t) - \vec{\mu}(\vec{x})\cdot\partial_x\rho(\vec{x},t)) + \partial_t(\vec{v}(\vec{x})\cdot\rho(\vec{x},t) - \vec{\mu}(\vec{x})\cdot\partial_x\rho(\vec{x},t)) \\
&= \partial_x\vec{v}(\vec{x})\rho(\vec{x},t) + \vec{v}(\vec{x})\partial_x\rho(\vec{x},t) - \partial_x\vec{\mu}(\vec{x})\partial_x\rho(\vec{x},t) - \partial_x^2\rho(\vec{\vec{x}},t)\vec{\mu}(\vec{x}) \\
&+ \vec{v}(\vec{x})\partial_t\rho(\vec{x},t) - \vec{\mu}(\vec{x})\partial_t\partial_x\rho(\vec{x},t)) \\
&= \partial_x\rho(\vec{x},t)(\vec{v}(\vec{x}) - \partial_x\vec{\mu}(\vec{x})) - \partial_x\vec{\mu}(\vec{x})\partial_x\rho(\vec{x},t) + \partial_x\vec{v}(\vec{x})\rho(\vec{x},t) + \vec{v}(\vec{x})\partial_t\rho(\vec{x},t) - \vec{\mu}(\vec{x})\partial_t\partial_x\rho(\vec{x},t))
\end{aligned} \tag{5}$$

For the change of ∂_x:

$$\partial_x\rho(\vec{x},t)(\vec{v}(\vec{x}) - \partial_x\vec{\mu}(\vec{x})) - \partial_x\vec{\mu}(\vec{x})\partial_x\rho(\vec{x},t) + \partial_x\vec{v}(\vec{x})\rho(\vec{x},t) \tag{6}$$

We see both $\vec{\mu}(\vec{x})$ and $\vec{v}(\vec{x})$ can change along with ∂_x. However, if we assume them as constants, then, they wil be 0. For the term $\partial_x\rho(\vec{x},t)(\vec{v}(\vec{x}) - \partial_x\vec{\mu}(\vec{x}))$, we can see how the velocity term can be interferenced by $\vec{\mu}(\vec{x})$, wheare as $\partial_x\vec{\mu}(\vec{x})$, is self-changing as well. For the change of ∂_t, the same story happens:

$$\vec{v}(\vec{x})\partial_t\rho(\vec{x},t) - \vec{\mu}(\vec{x})\partial_t\partial_x\rho(\vec{x},t)) \tag{7}$$

The change of $\rho(\vec{x},t)$ along with the $\vec{v}(\vec{x})$ direction can be interferenced by $\vec{\mu}(\vec{x})$, where its also second order (fasteer changing). However, if we ignore the term $-\vec{\mu}(\vec{x})\partial_t\partial_x\rho(\vec{x},t))$ and assume $\vec{\mu}(\vec{x})$ and $\vec{v}(\vec{x})$ as 0:

$$\vec{v}(\vec{x})\partial_t\rho(\vec{x},t)+\vec{v}(\vec{x})\partial_x\rho(\vec{x},t)=\vec{\mu}(\vec{x})\partial_x^2\rho(\vec{x},t)\Longrightarrow\vec{v}(\vec{x})(\partial_t\rho(\vec{x},t)+\partial_x\rho(\vec{x},t))=\vec{\mu}(\vec{x})\partial_x^2\rho(\vec{x},t) \tag{8}$$

With the proper mathematical notation, we can see that the advection is driven by $\vec{v}(\vec{x})$ and the diffusion process is driven by $\vec{\mu}(\vec{x})$. Since they're the terms on the left and right side of the equation, this indicates that the forces are balanced out, as well as a possible competition between the two. Now give the process a source term (come outside the domain Ω?), we have

$$\vec{v}(\vec{x})(\partial_t\rho(\vec{x},t)+\partial_x\rho(\vec{x},t))-\vec{\mu}(\vec{x})\partial_x^2\rho(\vec{x},t)=S(\vec{x})=\delta(\vec{x})=1 \tag{9}$$

The reason why we put $S(\vec{x})$ there is because it has to supply both of the processes, and $\delta(\vec{x})$ is a dirac delta fixed as 1. Now we give the advection process a constant a, a diffusion constant μ, and hide the vector forms:

$$\partial_t\rho(\vec{x},t)+a\partial_x\rho(\vec{x},t)-\mu\partial_x^2\rho(\vec{x},t)=S(\vec{x})\Longrightarrow\partial_t u+a\partial_x u=\mu\partial_x^2 u+S(x) \tag{10}$$

We have arrived the form from the assignment sheet. Now, we want to find the solution of our system. As we know:

$$\begin{aligned}\vec{F}(\vec{x},t)&=\vec{F_1}(\vec{x},t)+\vec{F_2}(\vec{x},t)=\vec{v}(\vec{x})\cdot\rho(\vec{x},t)-\vec{\mu}(\vec{x})\cdot\vec{\nabla}\rho(\vec{x},t)\\&=[\vec{v}(\vec{x})\cdot-\vec{\mu}(\vec{x})\cdot\vec{\nabla}]\rho(\vec{x},t)\end{aligned} \tag{11}$$

where $[\vec{v}(\vec{x})\cdot-\vec{\mu}(\vec{x})\cdot\vec{\nabla}]$ is our force operator, and $\vec{\nabla}=\partial_x$. Now assume $S(\vec{x})=0$, want to put our equation with an operator:

$$\vec{v}(\vec{x})\partial_t\rho(\vec{x},t)+[\vec{v}(\vec{x})\partial_x-\vec{\mu}(\vec{x})\partial_x^2]\rho(\vec{x},t)=0\Longrightarrow\partial_t\rho+[v\partial_x-\mu\partial_x^2]\rho=0 \tag{12}$$

This gives us the solution $\rho(\vec{x},t)=\rho_0(\vec{x}-\vec{v}t)$, which is an characteristic line, that is driven by the velocity term $\vec{v}$ from $\rho_0(\vec{x}-\vec{v}t)$ to $\rho(\vec{x},t)$. However, this process will be affected by the force operator defined ($[v\partial_x-\mu\partial_x^2]$), since we know $\dot{x}(t)$ gives $\vec{v}(\vec{x})$ (velocity) and $\ddot{x}(t)$ gives $\vec{a}(t)$ (acceleration). So $v\partial_x$ drives the process but $\mu\partial_x^2$ can accelerate the process.

1.2 How does the system change?

1.2.1 Fourier Mode

As we noticed, $S(\vec{x})=\delta(\vec{x})$ is our fundamental solution, which is bounded by 1. With the hint from lecture, we define our system in phase space as $Z(t)$, then we know that the divergence between the intial state and the final state can be expressed as:

$$|\delta Z(t)|\approx e^{\lambda t}|\delta Z_0| \tag{13}$$

where λ is the Lyapunov exponent, and δ is the change of the system. In our case, we have $e^{\mu t}$ and $1/\mu$ is Lyapunov time. This verified with our intuitive guess from 1.1 that μ is an interference factor. Moreover, we can analyze this through Fourier analysis. In 1D, we have:

$$\rho(\vec{x},t)=\int\hat{\rho}(k,\vec{x})e^{ik\vec{x}}dk \tag{14}$$

where k is our wavenumber, $e^{ik\vec{x}}=\cos(k\vec{x})+i\sin(k\vec{x})$, and:

$$\hat{\rho}(k,t)=\frac{1}{2\pi}\int e^{-ik\vec{x}}\rho(\vec{x},t)d\vec{x} \tag{15}$$

This shows, how the system switch from the spatial and phase domains. Moreover, as we remember, we have:

$$\partial_t\rho+[v\partial_x-\mu\partial_x^2]\rho=0\Longleftrightarrow\partial_t\rho=[-v\partial_x+\mu\partial_x^2]\rho \tag{16}$$

Now, we have time domain on the left-hand side, and spatial domain on the right-hand side. We define:

$$dt = \int \partial_t \hat{\rho}(k,t) e^{ik\vec{x}}, \;\; dx = \int \hat{\rho}(k,t)[-v\partial_x + \mu\partial_x^2] e^{ik\vec{x}} d\vec{x} \tag{17}$$

Moreover, $[-v\partial_x + \mu\partial_x^2]e^{ik\vec{x}} = (ivk - \mu k^2)e^{ik\vec{x}}$, and we should have:

$$\partial_t \rho = (ivk - \mu k^2)\hat{\rho}(k,t) \Longleftrightarrow \partial_t \rho = [-v\partial_x + \mu\partial_x^2]\rho(\vec{x},t) \tag{18}$$

We see how the force operator acts on our solution ρ in different domains. Now, continue with the fourier mode, we get:

$$\partial_t \rho = \int k e^{(ivk - \mu k^2)t} \rho_0(k) e^{ik\vec{x}} dk \tag{19}$$

We recognize that how the system would evolve with the time and phase domains. Interestingly, we also spot the Lyapunov exponent $e^{\mu t}$ in the time domain.

1.2.2 Numerical Solutions (DFT)

Up to now, we finished talking about the math and physics. We can start with the numerical solutions. In 1D, we have $x \in [0, L]$, where we discrete L into Δx with L/n, where n is the number of grid points. For our solution mesh, we then have:

$$u_j = u(x_j) = u(j\Delta x), \;\; u(x + L) = u(x) \tag{20}$$

In spatial and phase domains, we have:

$$u_j = u(x_j) = u(j\Delta x) \Longleftrightarrow u_j = e^{ikx_j} = e^{ik\Delta x_j} = e^{i\theta j} = e^{i(k \cdot L/n)j} \tag{21}$$

where $u = e^{ik\vec{x}}$ and $\theta = \frac{k \cdot L}{n}$, which gives $\theta = k\Delta x$, and k is the wavenumber again. For perodicity, we have $e^{i\theta(j+1) = e^{i\theta} e^{i\theta j}}$, where $j\Delta x = x_j$, in which j can be seen as n in $\Delta x = L/n$. So the number of grid points can affect how the system potentially oscillates, and $\Delta x \to 0 \Longleftrightarrow \theta \to 0$, indicating waves is related to spacing.

1.2.3 Numerical Solutions (Semi-discrete/CFL/Von Neumann stability)

Here, we will discuss the upwind, downwind and center schemes, and keep the force operators:

$$\hat{F}_p = [ivk - \mu k^2] \;\text{ acts on }\; \hat{\rho}(k,t) \Longleftrightarrow \hat{F}_s = [-v\partial_x + \mu\partial_x^2] \;\text{ acts on }\; \rho(\vec{x},t) \tag{22}$$

Where $\hat{F}_s$ and $\hat{F}_p$ means operators, and we start with the upwind scheme, with our PDE: $\partial_t \rho + \hat{F}_s \rho = 0$ for discretizations

$$\partial_x \rho \Longrightarrow \frac{1}{\Delta x}[\rho(x_j,t) - \rho(x_{j-1},t)] \tag{23}$$

We then have:

$$\partial_t \rho(t) = -\frac{\hat{F}_s}{\Delta x}(\rho_j - \rho_{j-1}) \Longleftrightarrow \partial_t \hat{\rho}(t) = -\frac{\hat{F}_p}{\Delta x}(1 - e^{-i\theta})\hat{\rho}(t) \tag{24}$$

with the assumption that $\rho_{j+n} = \rho_j$, and $\rho_j(t) = \sum e^{i\theta j}\hat{\rho}_j(t)$. Also note that $\theta = k\Delta x$, this means:

$$\partial_t \hat{\rho}(t) = -\frac{\hat{F}_p}{\Delta x}(1 - e^{-i\theta})\hat{\rho}(t) = -\frac{\hat{F}_p}{\Delta x}(1 - e^{-ik\Delta x})\hat{\rho}_j(t) \tag{25}$$

If Δx is small, we can Taylor expand it to calculate the errors up to $\mathcal{O}(\Delta x^2)$. Then, we achieve:

$$\hat{\rho}_j(t) = \hat{\rho}(0) e^{-ikvt} \cdot e^{-v\Delta x k^2 t/2} \tag{26}$$

where $\mu = v/2\Delta x$, and $e^{-v\Delta x k^2 t/2}$ is the correlation term, it runs off the errors, but it smooths everything out, and e^{-ikvt} is the advection term. For the downwind scheme, we have:

$$\partial_t \rho_j = -\frac{\hat{F_s}}{\Delta x}(\rho_{j+1} - \rho_j) \Longleftrightarrow \hat{\rho}_j(t) = \hat{\rho}(0) e^{-ikvt} \cdot e^{v\Delta x k^2 t/2} \tag{27}$$

However, the system is not stable because of the correlation term, with the positive sign. Now, we have the center scheme:

$$\partial_t \rho_j = -\frac{\hat{F_s}}{2\Delta x}(\rho_{j+1} - \rho_{j-1}) \Longrightarrow \rho_{j_{m+1}} = \rho_{j_m} - \frac{\Delta t \hat{F_s}}{2\Delta x}(\rho_{j+1,m} - \rho_{j-1,m}) \tag{28}$$

with $\rho_{jm} = \hat{\rho}_m e^{i\theta j}$, and $\rho_{j+1,m} = e^{i\theta}$ and $\rho_{j-1,m} = e^{-i\theta}$, we have:

$$\hat{\rho}_{m+1} = \hat{\rho}_m - \frac{\Delta t \hat{F_s}}{2\Delta x}(e^{i\theta} - e^{-i\theta})\hat{\rho}_m \Longleftrightarrow \hat{\rho}_{m+1} = m(\theta)\hat{\rho}_m \tag{29}$$

where $m(\theta) = 1 - \frac{\Delta t \hat{F_s}}{2\Delta x}(e^{i\theta} - e^{-i\theta})$, which is the amplifier. With $|m(\theta)| > 1$, unstable and $|m(\theta)| < 1$, stable. For Von Neumann stability, we have L^2 stable, if $max_\theta |m(\theta)| \leq 1$ and $max_\theta |m(\theta)| > 1$ for unstable condition, with:

$$MV(\theta) = m(\theta)V(\theta) \tag{30}$$

where $m(\theta)$ is the symbol of the scheme and $V(\theta) = e^{i\theta j}$. Ideally, we can check this stability with a unit circle, to see whether the solution remains in it or not.

1.2.4 LAX equivalent Thm

We have a PDE $\partial_t u = LU$ with $u(x,0) = U_0(x)$, and suppose it is L^2 stable, then we have:

$$\int u^2(\vec{x},t)d\vec{x} \leq C_t \int u^2(\vec{x},0)d\vec{x} \tag{31}$$

which gives $U_j^{(k)} \cong U(x_j, tk)$, where $U^{(k)}$ is a grid vector at time tk grid function. Then we have:

$$\|U^{(k)}\|_2 \leq C_t \|U^{(0)}\|_2 \Longleftrightarrow U^{(k-1)} = MU^{(k)}, \text{ if } tk \leq k \tag{32}$$

Then, according to LAX equivalent theorem, we have Error $= \mathcal{O}(\Delta t^p, \Delta x^p)$, where p is the truncation error order. Moreover, we have $\Delta t \cong \Delta x^2$ for diffusion and $\Delta t \sim \Delta x$ for advection.

2 Exercise 1

2.1 Suppose that S is the grid function $S_j = S(x_j)$ and $u(x,t)$ is a smooth function of x and t, and $U_j(0) = u(xj, 0)$. Compute the symbol of the right side of (2). In other words, define the translation invariant operator $M = -D_0 + \mu D_+ D_-$ and compute $m(\theta)$

Proof. We have the equation:

$$\begin{aligned} \dot{U} = -D_0 U + \mu D_+ D_- U + S &\iff \dot{U}(\vec{x}_j, tk) = -D_0 U\vec{x}_j, tk) + \mu(\vec{x}_j)(D_+ D_- U(\vec{x}_j, tk)) + S(\vec{x}_j) \\ &\iff U_j^{(k+1)} - U_j^{(k)} = (\mu D_+ D_- U_j^{(k)} - D_0 U_j^{(k)}) + S_j)\Delta t \end{aligned} \tag{33}$$

Since $S_j = 1$, $S_j \Delta t = 0$. We have:

$$U_j^{(k+1)} = U_j^{(k)} + \Delta t[(-D_0 + \mu D_+ D_-)U_j^{(k)}] \Longrightarrow M = I + (-D_0 + \mu D_+ D_-)\Delta t \tag{34}$$

Because of $U^{(k+1)} = MU^{(k)}$. Moreover, for $MV(\theta)$, we have:

$$V_j + \left[-\frac{\Delta t}{2\Delta x}(V_{j+1} - V_{j-1}) + \frac{\mu \Delta t}{\Delta x^2}(V_{j+1} - 2V_j + V_{j-1}) \right] \tag{35}$$

Since $V_j = e^{i\theta j}$, we also have:

$$\begin{aligned} & e^{i\theta j} + \left[-\frac{\Delta t}{2\Delta x}(e^{i\theta(j+1)} - e^{i\theta(j-1)}) + \frac{\mu \Delta t}{\Delta x^2}(e^{i\theta(j+1)} - 2e^{i\theta j} + e^{-i\theta(j-1)}) \right] \\ \Longrightarrow & e^{i\theta j} + \left[-\frac{\Delta t}{2\Delta x}(e^{i\theta} - e^{-i\theta})e^{i\theta j} + \frac{\mu \Delta t}{\Delta x^2}(e^{i\theta} - 2 + e^{-i\theta})e^{i\theta j} \right] \\ \Longrightarrow & e^{i\theta j} + \left[-\frac{i\Delta t}{\Delta x}\sin(\theta)e^{i\theta j} + \frac{\mu \Delta t}{\Delta x^2}(\cos(\theta) - 1)e^{i\theta j} \right] \\ \Longrightarrow & \left[1 - \frac{i\Delta t}{\Delta x}\sin(\theta) + \frac{2\mu \Delta t}{\Delta x^2}(\cos(\theta) - 1) \right] e^{i\theta j} = m(\theta)e^{i\theta j} \end{aligned} \tag{36}$$

Thus, we have:

$$m(\theta) = 1 - \frac{i\Delta t}{\Delta x}\sin(\theta) + \frac{2\mu \Delta t}{\Delta x^2}(\cos(\theta) - 1) \tag{37}$$

□

2.2 Use the result to show the semi-discrete scheme is von-Neumann stable.

Proof. We want to show that $\max_\theta |m(\theta)| \leq 1$ for von-Neumann stablility, and we have:

$$\begin{aligned} m(\theta) &= 1 - \frac{i\Delta t}{\Delta x}\sin(\theta) + \frac{2\mu \Delta t}{\Delta x^2}(\cos(\theta) - 1) \\ &= 1 - \Delta t \left[\frac{i}{\Delta x}\sin(\theta) - \frac{2\mu}{\Delta x^2}(\cos(\theta) - 1) \right] \end{aligned} \tag{38}$$

We want to show the terms after 1 is < 1, define our system as:

$$\begin{aligned} x(t) &= A(\cos(\theta) - 1)\Delta t + B\sin(\theta)\Delta t \\ &= \mathrm{Re}Ce^{i(\theta+\delta)t} = |C|\cos(\theta + \delta) = \sqrt{A^2 + B^2}\cos(\theta + \delta) \end{aligned} \tag{39}$$

where δ indicates small changes, and C is our damping factor, and $A = -2\mu/\Delta x^2$ and $B = i/\Delta x$, and $C = f(\mu, \Delta x) = \sqrt{(4\mu^2 - \Delta x^2)/\Delta x^4} \leq \mathcal{O}(1)$. Since $\cos(\theta + \delta) \leq 1$ for all $\theta + \delta$, this means that $\mathrm{Re}Ce^{i(\theta+\delta)t} \leq 1$,

and $\max_\theta |m(\theta)| \leq 1$. Hence, the semi-discrete scheme is von-Neumann stable. Now, we want to find $\max_\theta |m(\theta)|$ and $\min_\theta |m(\theta)|$:

$$\begin{aligned} |m(\theta)|^2 &= \left(\frac{2\mu\Delta t}{\Delta x^2}(\cos(\theta)-1)\right)^2 + \left(1 - \frac{i\Delta t}{\Delta x}\sin(\theta)\right)^2 \\ &= \frac{4\mu^2\Delta t}{\Delta x^2}(\cos(\theta)-1)^2 + \left(1 + \frac{\Delta t}{\Delta x}\sin^2(\theta)\right) \end{aligned} \tag{40}$$

To find the critical points for $|m(\theta)|^2$, differentiate with respect to θ

$$\begin{aligned} \frac{d}{d\theta}|m(\theta)|^2 &= -\frac{8\mu^2\Delta t}{\Delta x^2}(\cos(\theta)-1)\sin(\theta) + \frac{2\Delta t}{\Delta x}\sin(\theta)\cos(\theta) \\ &= \sin(\theta)\left(-\frac{8\mu^2\Delta t}{\Delta x^2}(\cos(\theta)-1) + \frac{2\Delta t}{\Delta x}\cos(\theta)\right) = 0 \end{aligned} \tag{41}$$

This yields critical points from $\sin(\theta) = 0$, which gives us $\theta = 0, \pi, 2\pi$, $\theta \in [0, 2\pi]$. We can also achieve 0 inside the bracket with $\cos(\theta)$ but that depends on the values of Δx and μ also. Now, we want to evaluate $|m(\theta)|$ at boundary points:

$$|m(\theta)| = \begin{cases} 0 & \text{for } \theta = 0, \text{ and } \theta = 2\pi, \text{local minima} \\ 1 - (2\mu + \Delta x)\Delta t/\Delta x^2 & \text{for } \theta = \pi/2, \text{local maximum depending on } \mu \text{ and } \Delta x \text{ and } \Delta t \\ 1 - 4\mu\Delta t/\Delta x^2 & \text{for } \pi, \text{local maximum depending on } \mu \text{ and } \Delta x \text{ and } \Delta t \\ 1 + (2\mu - \Delta x)\Delta t/\Delta x^2 & \text{for } \theta = 3\pi/2, \text{local maximum depending on } \mu \text{ and } \Delta x \text{ and } \Delta t \end{cases}$$

which gives us local minimum and maximum points. □

2.3 Show that the computed solution is second order accurate in the sense that, for any fixed t

$$\Delta x \sum_{j=1}^{n} [u(x_j, t) - U_j(t)]^2 = \mathcal{O}(\Delta x^4)$$

Proof. We compute the Taylor expansions of $u(x_{j+1}, t)$ and $u(x_{j-1}, t)$ around x_j first

$$u(x_{j+1}, t) = u(x_j, t) + u_x(x_j, t)\Delta x + \frac{1}{2}u_{xx}(x_j, t)\Delta x^2 + \frac{1}{6}u_{xxx}(x_j, t)\Delta x^3 + \mathcal{O}(\Delta x^4) \tag{42}$$

$$u(x_{j-1}, t) = u(x_j, t) - u_x(x_j, t)\Delta x + \frac{1}{2}u_{xx}(x_j, t)\Delta x^2 - \frac{1}{6}u_{xxx}(x_j, t)\Delta x^3 + \mathcal{O}(\Delta x^4) \tag{43}$$

Then we want to estimate the truncation errors introduced by the difference operators D_0 and D_+D_-

$$\begin{aligned} D_0 u_j &= \frac{1}{2\Delta x}(u(x_{j+1}, t) - u(x_{j-1}, t)) \\ &= u_x(x_j, t) + \frac{1}{6}u_{xxx}(x_j, t)\Delta x^2 + \mathcal{O}(\Delta x^3) \end{aligned} \tag{44}$$

Then we also have

$$\begin{aligned} D_+D_- u_j &= \frac{1}{\Delta x^2}(u(x_{j+1}, t) - 2u(x_j, t) + u(x_{j-1}, t)) \\ &= u_{xx}(x_j, t) + \mathcal{O}(\Delta x^2) \end{aligned} \tag{45}$$

The truncation error for $D_0 u_j$ and $D_+D_- u_j$ are thus $\mathcal{O}(\Delta x^2)$. Then we have

$$\begin{aligned} \Delta x \sum_{j=1}^{n} [u(x_j, t) - U_j(t)]^2 &= \Delta x \sum_{j=1}^{n} \left[\mathcal{O}(\Delta x^2)\right]^2 \\ &= n\Delta x \mathcal{O}(\Delta x^4) = \mathcal{O}(\Delta x^4) \end{aligned} \tag{46}$$

where $n\Delta x = L$. Hence, we have a second order accuracy. □

3 Exercise 2

3.1 Code the forward Euler scheme (4). Suppose μ is small (say $\mu = 0.1$ or $\mu = 0.01$). Do numerical experiments (plot the numerical solution) on a variety of meshes up to time $t = 0.5$ and several values of Δt to see whether the put diffusion CFL condition (5) determines the stability.

Proof. Define our whole system as BL $= f(\Delta x, \Delta t, \mu, n, L, S, U, M)$, where M is an operator, $M = -D_0 + \mu D_+ D_-$ that acts on our solution (U). We keep $S = 1, L = 1$, and exam $\mu = 1, \mu = 0.1$ and 0.01, then we tune the numerical parameters $\Delta x, \Delta t$ and n. We decided to fix $n = [50, 100, 200]$, and we also know

$$\Delta x = \frac{L}{n+1}, \quad \Delta t \leq \frac{0.5\Delta x^2}{\mu} \quad \text{(CFL stability)} \tag{47}$$

which tell us CFL $= f(\Delta t, \Delta x(n, L), \mu)$. So, we can set up an upper bound for Δt for stability purpose that automatically calculates the right Δt for us based on CFL. Then, we should have the correct solution as:

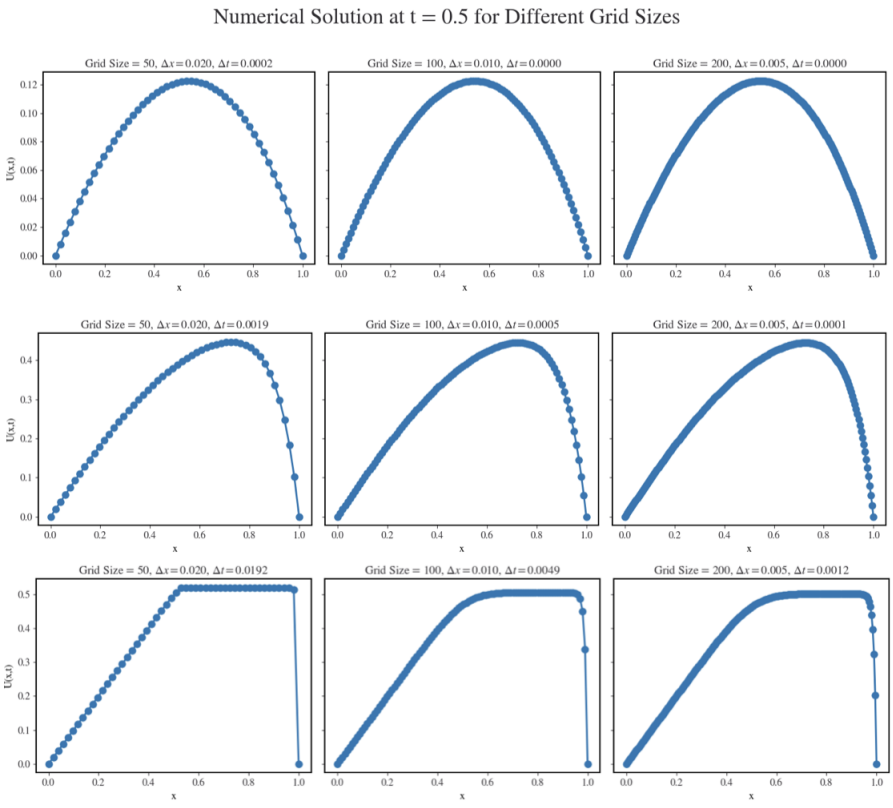


Figure 1: The stable solutions $U(x, t)$ that safeties the CFL condition at $t = 0.5$ with different grid points $n = 50, 100, 200$. **Top row** with $\mu = 1$, **Middle row** with $\mu = 0.1$; **Bottom row** with $\mu = 0.01$

Those processes are stable by the way we calculated them. Moreover, with dirichlet boundary condition $u(0, t) = 0,\ u(L, t) = 0$ and initial condition $u(x, 0) = 0,\ 0 < x < L$, we can see how the advection diffusion process changes with Δt between the spatial bound $0 < u < L = 1$(x-axis) and the time bound $0 < t < 0.5$ (y-axis). We also noticed that the lines are smoother, as n progress bigger due to the resolution change of the grid, whereas lower resolution can give numerical artifacts (we should not have a corner edge). Moreover, at $x = 1.0$(L), it seems to be a boundary layer, and it seems to be more propounded with $\mu = 0.001$ (when μ is smaller):

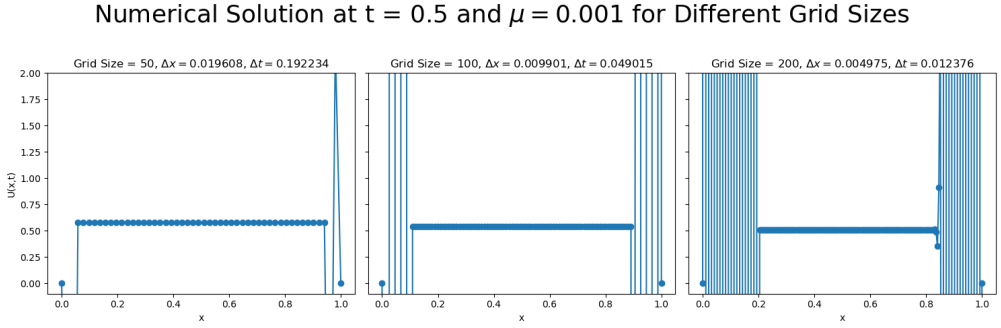


Figure 2: The "boundary layer" at $x = 0$ and $x = L$.

Lastly, there are also some experimental results that we can show in which they don't satisfies CFL conditionbut are interesting to see:

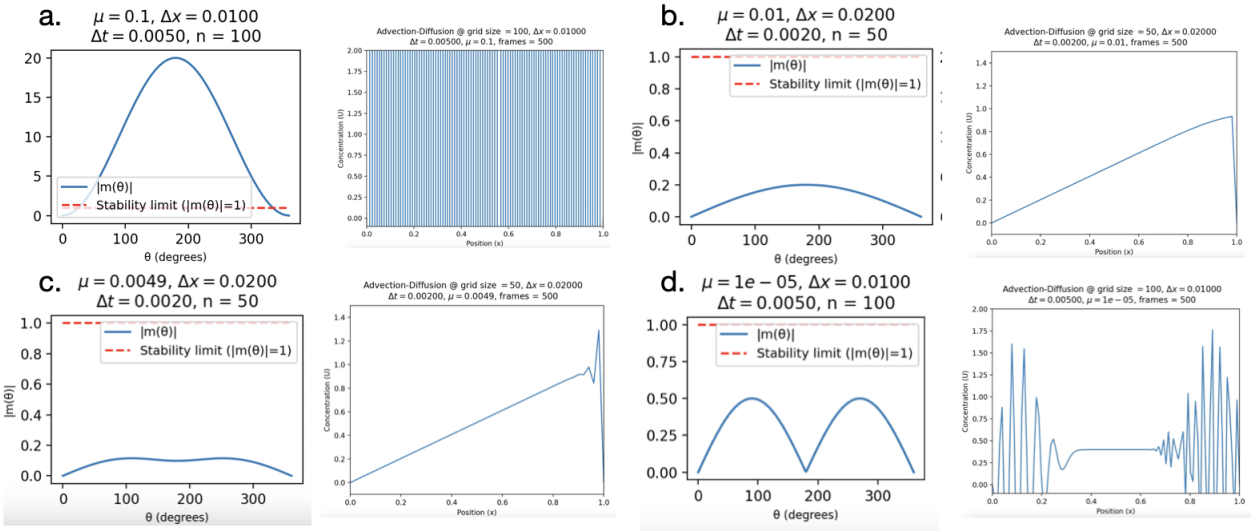


Figure 3: Experimental pictures with different μ, Δx, Δt conditions.

Discussions: **a** gives us an unstable result, **b** gives us a stable result, **c** gives us a stable result but with some oscillating behaviors observed at the boundary $x = L$, and lastly, **d** also gives us a stable result but with some oscillation behaviors at both boundary ends $x = 0$ and $x = L$. Those strange behaviors can be explained by the next subsection. However, with all the evidences we just discussed, we now understand the importance of the CFL condition, but also, at the same time, those oscillating behaviors, reminds us the damping factor from the previous section: $C = \sqrt{(4\mu^2 - \Delta x^2)/\Delta x^4} \leq \mathcal{O}(1)$ is also important, as well as the right μ, Δt and Δx values. □

3.2 Make plots of $|m(\theta)|$ for $0 \leq \theta \leq 2\pi$ to see whether the numerical stability/instability you see in the code agrees with the stability prediction based on max $|m(\theta)|$.

Proof. In the previous section, we have shown that the stability related to the local minima and maxima areas are:

$$|m(\theta)| = \begin{cases} 0 & \text{for } \theta = 0, \text{ and } \theta = 2\pi, \text{local minima} \\ 1 - (2\mu + \Delta x)\Delta t/\Delta x^2 & \text{for } \theta = \pi/2, \text{local maxima depending on } \mu \text{ and } \Delta x \text{ and } \Delta t \\ 1 - 4\mu\Delta t/\Delta x^2 & \text{for } \pi, \text{local maxima depending on } \mu \text{ and } \Delta x \text{ and } \Delta t \\ 1 + (2\mu - \Delta x)\Delta t/\Delta x^2 & \text{for } \theta = 3\pi/2, \text{local maxima depending on } \mu \text{ and } \Delta x \text{ and } \Delta t \end{cases}$$

So, we have local minima at $\theta = 0, 2\pi$, and local maxima at $\theta = \pi/2, \pi, 3\pi/2$ based on $\mu, \Delta x, \Delta t$.

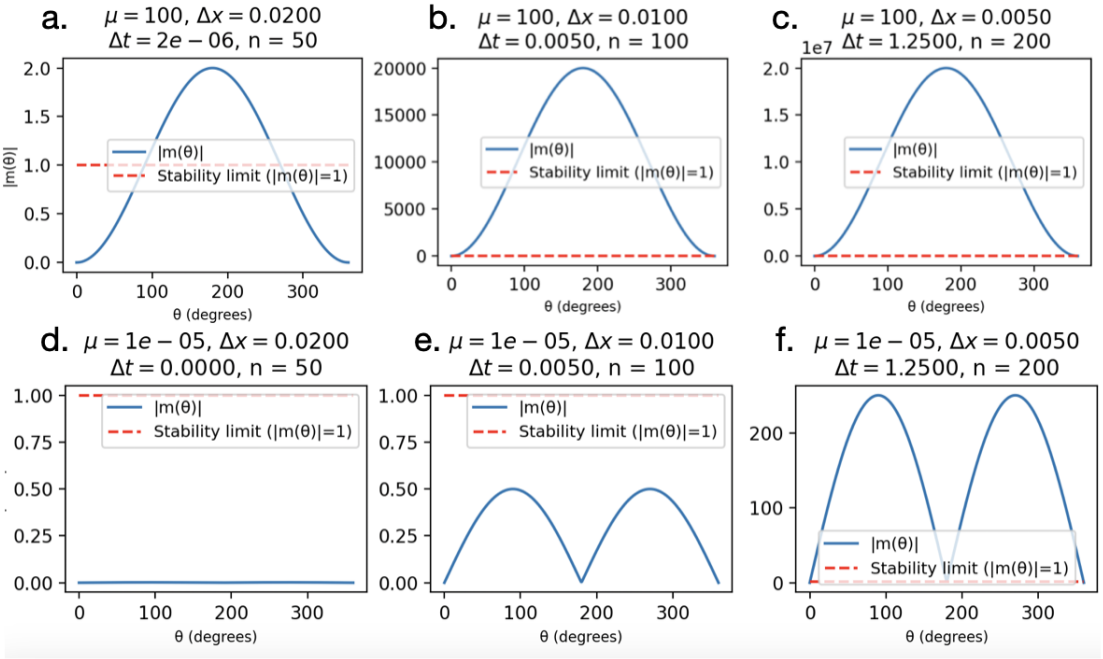


Figure 4: $|m(\theta)|$ over $0° \leq \theta \leq 360°$ with an arbitrary selection of $\Delta x = 0.02, 0.01, 0.005$ and very large $\mu = 100$ and very small $\mu = 0.00001$. Plots are **a, b, c, d, e, f**, each with different condition variations

As we can see from the plots, there are majorly two shapes we achieved: 1) one bell and 2) two bells. Those plots show exactly, what we have calculated from the table. The local minima occurs at the $\theta = 0° = 360°$, which can be sen from all the plots, and the local maxima varies with $\mu, \Delta x, \Delta t$ at $\theta = 90°$, $\theta = 270°$ and $\theta = 180°$. Now, we can further evaluate the correctness of our table with some numbers plug in:

a. local maxima at $\theta = 180° : |m(\theta)| = 1$

b. local maxima at $\theta = 180° : |m(\theta)| = |1 - 20000| > 1$

c. local maxima at $\theta = 180° : |m(\theta)| = |1 - 2e7| > 1$

e. local maxima at $\theta = 90° : |m(\theta)| = |1 - 0.5|$, $\theta = 270°, |m(\theta)| = |1 + (-0.499)| < 1$

f. local maxima at $\theta = 90° : |m(\theta)| = |1 - 251|$, $\theta = 270°, |m(\theta)| = |1 + (-249)| > 1$

We can see that our calculations agree with our **b, c, e, f**, but not **a**. This can be explained by the numerical artifact, where the floating point errors accumulate and machine Epsilon play in, since our time step is very small which is $\Delta t = 2e - 0.6$ and $t = 0.5$. So we have lots of incrementing here. In iterative methods like the Forward Euler method, this can lead significant errors, which is something we need to be mindful with. However, overall, the numerical stability/instability we see from the plots, agree with our predictions on $\max|m(\theta)|$.

In terms of the reasons for those stable and unstable behaviors, we can start with talking about μ. As we can see, we picked a large $\mu = 100$ and a small $\mu = 1e - 05$. This is because μ is our Lyapunov exponent $e^{\mu t}$, and $|\delta Z(t)| \approx e^{\lambda t}|\delta Z_0|$. It takes care of the stability. Moreover, we also know that it appears in our fourier mode as well. In ths phase domain, we have our PDE evolves as:

$$\partial_t \rho = \int k e^{(ivk - \mu k^2)t} \rho_0(k) e^{ik\vec{x}} dk \tag{48}$$

This equation tells us the growths and decay of our wave propagation in the phase space, where ivk leads to oscillatory behavior (imaginary exponential), and$-\mu k^2$ contributes to the exponential decay of the amplitude of the components. So, when we have a small μ, within $|m(\theta)|$, we can have oscillations observed (as mentioned in the previous section). This is because a smaller μ means that the damping effect is weaker, allowing the high frequency components to persist longer. This persistence can manifest as oscillations. Also, since μ is small, ivk will most likely to be dominated, leading to more pronounced oscillatory behavior. This also agrees with our plots. We have possible infinite oscillations at $\theta = 90°$ and $\theta = 270°$ due to the potential energy landscape we observe from **e, f**.

Now, with a large μ, it means reducing the need to handle high-frequency oscillations or instabilities. Since it will be dominated over the imaginary component. This also match our observations from **b, c**, where we only have one local maxima at $\theta = 180°$. However, if $|m(\theta)| > 1$, we will still have instability regardless. □

4 Exercise 3

4.1 Use good Δx and Δt values from Exercise 2 to make a movie of an accurate simulation of (1) up to the time the solution seems to have settled into a steady state. Describe the time dependent and steady state behavior when μ is small. There will be a boundary layer either near $x = 0$ or near $x = L$. Most of the behavior may be explained using the simplification $\mu = 0$, but not the boundary layer.

Proof. We take the good $\Delta t = 0.00005$ and $\Delta x = 0.0099$ from exercise 2, we did three movies with $\mu = 1$, $\mu = 0.1$ and $\mu = 0.01$, and the plots can be seen below, where the movies can be found in the attachment:

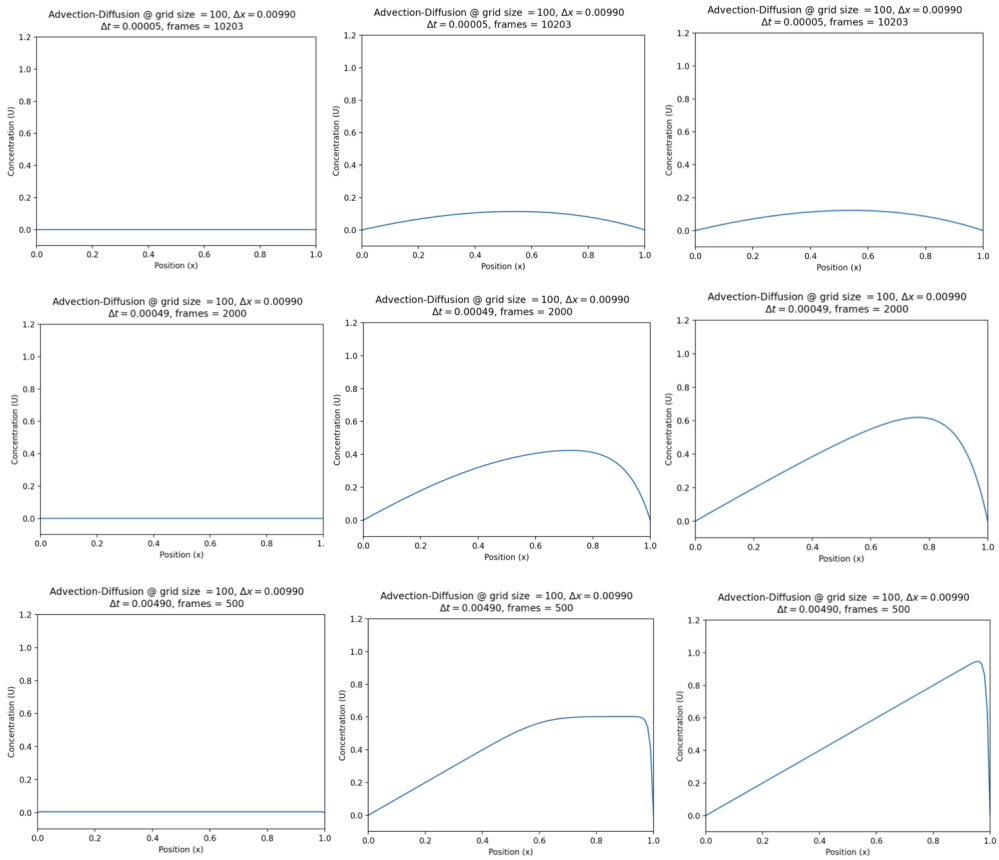


Figure 5: Advection-Diffusion simulations at grid size = 100, with $\Delta x = 0.0099, \Delta t = 0.00005$ with diffusion coefficients $\mu = 1$(top row), $\mu = 0.1$, $\mu = 0.01$(bottom row). **Left:** time frame at the beginning; **Middle:** time frame in the middle; **Right:** time frame at the end.

As we know, we have CFL as:

$$\text{CFL} = \frac{\mu \Delta t}{\Delta x^2} \leq 0.5, \quad \Delta t \leq \frac{0.5 \Delta x^2}{\mu} (\text{CFL stability}) \tag{49}$$

As we recognized from the CFL stability expression, $\Delta t = f(C_{max}, \Delta x, \mu)$, where C_{max} is the Courant's constant, here as 0.5. So, Δt is inversely proportional to μ but proportional to Δx. Keep Δx fixed, when we have a small μ, Δt will have a lager stability bound, which leads to steady state more easily. Moreover, as we learned from section (1.1), the motion of the characteristic line can be affected by the force operator $\hat{F}_s = v\partial_x - \mu\partial_x^2$, where μ affects the term that is second order with acceleration to decrease the velocity of

the characteristic line. So, when we have a smaller μ, we most likely will move faster and reach out to the steady state quickly. However, it doesn't change the fact that we have a boundary layer at $x = 0$ or $x = L$. This could possibly related to the numerical artifacts. Since we understand those might be related to Δx, we want to do some plotting investigations first:

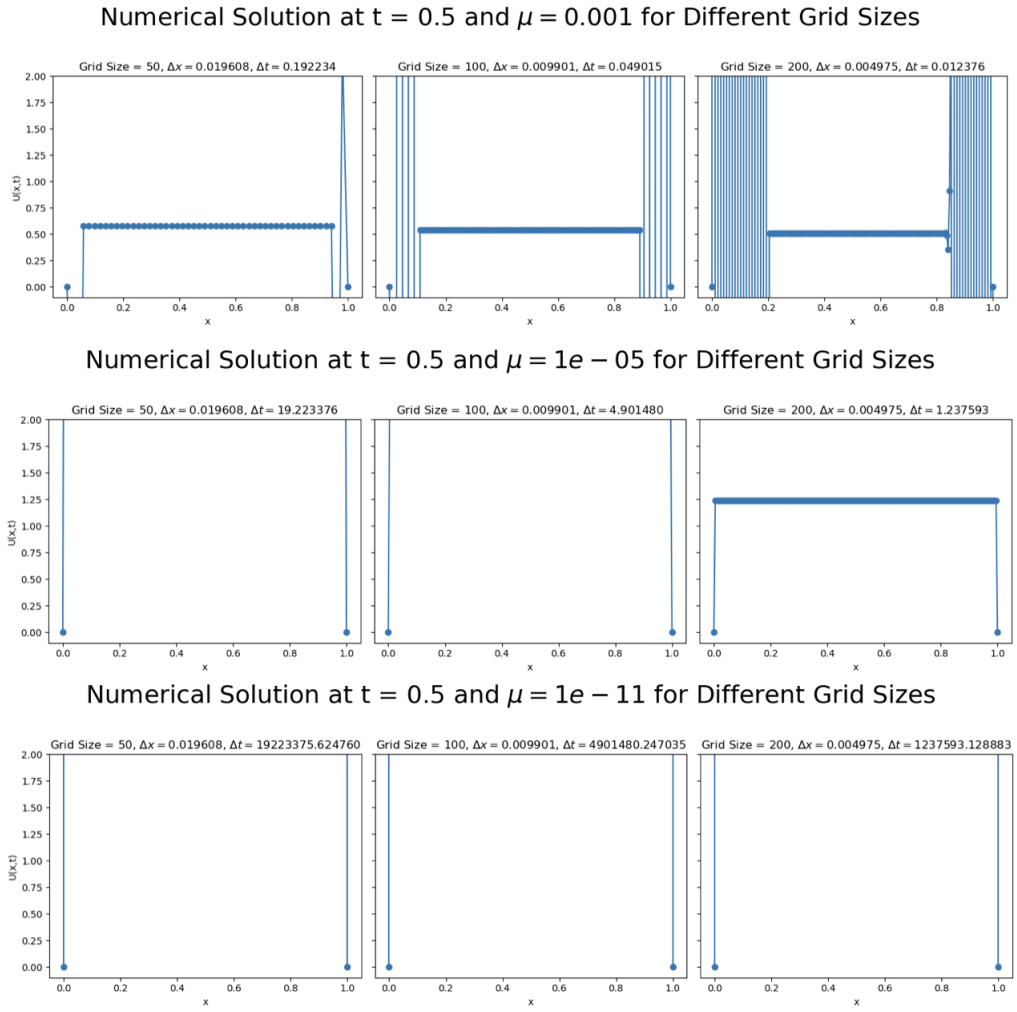


Figure 6: Boundary Layer examinations at $\mu = 0.001$, $\mu = 1e-05$, $\mu = 1e-11$. **Left:** pictures at $n = 50$; **Middle:** pictures at $n = 100$; **Right:** pictures at $n = 200$

As we can, the boundary layer effect can be minimized when we reduce μ, and have a bigger n (small Δx), but it will not go away. Now, we want to see what happens when $\mu = 0$. This means that we change our PDE and CFL condition into:

$$\partial_t u + a\partial_x u = \mu u_x^2 + S(x) \Longrightarrow \partial_t u + a\partial_x u = S(x), \quad \mu\frac{\Delta t}{\Delta x^2} \leq \frac{1}{2} \Longrightarrow \frac{\Delta t}{\Delta x} \leq \frac{1}{2}, \quad \Delta t \leq 0.5\Delta x (\text{CFL stability}) \tag{50}$$

We have to change CFL condition too if $\mu = 0$. Because our original CFL won't be true and it will be hard to find the real solution if CFL doesn't hold. Now, we can see the $\mu = 0$ case below:

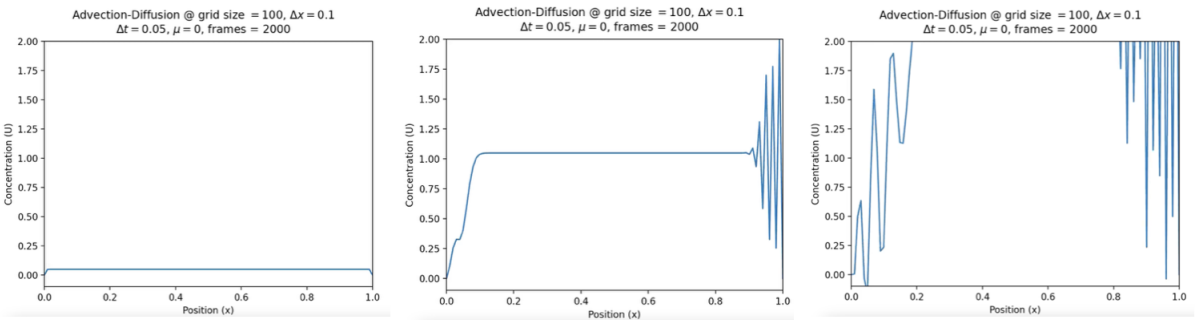


Figure 7: Advection process at $n = 100$, $\Delta x = 0.1$, $\Delta t = 0.05$. **Left:** beginning process; **Middle:** middle process (close to the beginning); **Right:** intermediate process

.

An avection process, in this problem, will be a line moving from $x = 0$ to $x = L$, with the change of its concentration $U(\vec{x})$. Hence it doesn't not only vary with the $x - axis$ $(\vec{x})$ but also the $y - axis(U(\vec{x}))$. So it will be the characteristic line as we previously discussed. However, here we don't have the diffusion process. Hence:

$$\vec{v}(\vec{x})\partial_t\rho(\vec{x},t) + [\vec{v}(\vec{x})\partial_x]\rho(\vec{x},t) = 0 \Longrightarrow \partial_t\rho + [v\partial_x]\rho = 0 \tag{51}$$

where our force operators are reduced to $\hat{F}_s = [\vec{v}(\vec{x})]$ (spatial domain) $\Longleftrightarrow$ $\hat{F}_p = [ivk]$ (phase domain). In terms of the solution of our PDE, we have: $\rho(\vec{x},t) = \rho_0(\vec{x} - \vec{v}t)$, which is a characterstic line, where $\vec{v} = \hat{F}_s$. This gives us what happens from **Left** and **Middle**. For the oscillation part, is because we reduced $\hat{F}_p = [ivk - \mu k^2] \Longrightarrow \hat{F}_p = [ivk]$. Without the real part of the $\hat{F}_p$, we are more easy to get oscillations. However, having a advection process only doesn't change the fact that we have numerical boundary layers at $x = 0$ and $x = L$ (**Right**). □

5 Exercise 4

5.1 Consider the partly implicit scheme

$$U^{(k+1)} = U^{(k)} + \Delta t\left[D_0U^{(k)} + \mu D_+D_-U^{(k+1)} + S_j\right]$$

Show that the scheme is first order accurate if $\Delta t = \Delta x$ and $\mu = 1$.This means showing the residual has the appropriate size and the scheme is stable.

Proof. Consider S_j as a grid function, $S_j \Longleftrightarrow S(\vec{x_j})$, and we substitute $\Delta t = \Delta x$ and $\mu = 1$, we have:

$$\begin{aligned}
&U^{(k+1)} - U^{(k)} = \Delta x\left[D_0U^{(k)} + D_+D_-U^{(k+1)} + S_j\right] \\
&\Longleftrightarrow U^{(k+1)} - \Delta xD_+D_-U^{(k+1)} = U^{(k)} + \Delta xD_0U^{(k)} + \Delta xS_j \\
&\Longleftrightarrow [1 - \Delta xD_+D_-]U^{(k+1)} = [1 + \Delta xD_0]U^{(k)} + \Delta xS_j \\
&\Longleftrightarrow [1 - \Delta xD_+D_-]U^{(k+1)} - [1 + \Delta xD_0]U^{(k)} = \Delta xS_j
\end{aligned} \tag{52}$$

where the left hand side is our numerical iteration with incrementing in k, and right hand side is our residual. $[1 - \Delta xD_+D_-]$ is an operator acts on $U^{(k+1)}$ and $[1 + \Delta xD_0]$ is an operator acts on $U^{(k)}$. Clearly $\Delta xS_j \approx \mathcal{O}(\Delta x)$, where $S_j \in [0, L]$ and $L = 1$. Hence its first order accurate. □

Proof. We can also do this more rigorously by considering a Fourier mode $u(k) = r^k \cdot e^{i\theta j}$ that gives us homogeneous partly implicit scheme

$$r^{k+1}e^{i\theta j} = r^ke^{i\theta j} + \frac{\Delta t}{2\Delta x}\left(r^ke^{i\theta(j+1)} - r^ke^{i\theta(j-1)}\right) + \mu\frac{1}{\Delta x^2}\left(r^{k+1}e^{i\theta(j+1)} - 2r^{k+1}e^{i\theta j} + r^{k+1}e^{i\theta(j-1)}\right) \tag{53}$$

with $\mu = 1$ and $\Delta t = \Delta x$, we have

$$r = 1 + \frac{\Delta x}{2\Delta x}\left(e^{i\theta} - e^{-i\theta}\right) + \frac{1}{\Delta x^2}\left(re^{-2i\theta} - 2r + re^{2i\theta}\right) \Longrightarrow |r|^2 = \frac{1 + \sin^2\theta}{1 + 4\frac{1}{\Delta x}(1 - \cos\theta) + \frac{4}{\Delta x^2}(1 - \cos\theta)^2} \tag{54}$$

Hence we should have $0 \leq \sin^2\theta \in \mathcal{O}(1)$, which gives us

$$0 \leq \frac{4}{\Delta x}(1 - \cos(\theta)) + \frac{4}{\Delta x^2}(1 - \cos(\theta))^2 \tag{55}$$

This indicates that $\leq \mathcal{O}(\frac{1}{\Delta x^2}) \approx \mathcal{O}(1)$. Hence we have $|r|^2 \leq 1$, $|r| \leq 1$, and our solution is first order accurate. □

Proof. Now, to do Von Neumann stability analysis, we know:

$$D_0 V_j = \frac{i}{\Delta x}\sin(\theta)e^{i\theta j}, \quad D_+ D_- V_j = \frac{2}{\Delta x^2}(\cos(\theta) - 1)e^{i\theta j} \tag{56}$$

With $V_j^{(k)} = e^{i\theta j}$, we assume an amplification factor $m(\theta)$, such that the next time step $V_j^{(k+1)} = m(\theta)e^{i\theta j}$. Moreover, the source S_j is a grid function (same as V_j), so $S_j e^{i\theta j}$, and

$$m(\theta)e^{i\theta j} - e^{i\theta j} = \Delta x\left[\frac{i}{\Delta x}\sin(\theta)e^{i\theta j} + \frac{2}{\Delta x^2}(\cos(\theta) - 1)m(\theta)e^{i\theta j} + S_j e^{i\theta j}\right] \tag{57}$$

$$\Longrightarrow m(\theta) = \frac{1 + i\sin(\theta)}{1 - \frac{2}{\Delta x}(\cos(\theta) - 1)} = \frac{1 + \sin^2(\theta)}{\left(1 + \frac{2}{\Delta x}(1 - \cos(\theta)\right)^2} = \frac{1 + \sin^2(\theta)}{1 + \frac{4(1-\cos(\theta))}{\Delta x} + \frac{4(1-\cos(\theta))^2}{\Delta x^2}} \tag{58}$$

Now, we compare $\sin^2(\theta)$ and $4(1 - \cos(\theta))^2$, and we have $1 - t^2 < 4(1 - t)^2, -t < t < 1$. Hence we have $|\cos(\theta)| \leq 1$, which means $|m(\theta)| \leq 1$. So our scheme is stable. □

6 Exercise 5

6.1 Code a tri-diagonal solver to find V that satisfies

$$(I - cD_+D_-)V = F$$

Use this to implement the partly implicit scheme.See whether you can do the computation of Exercise 3 faster, first for $\mu = 1$ and then for smaller values of μ.

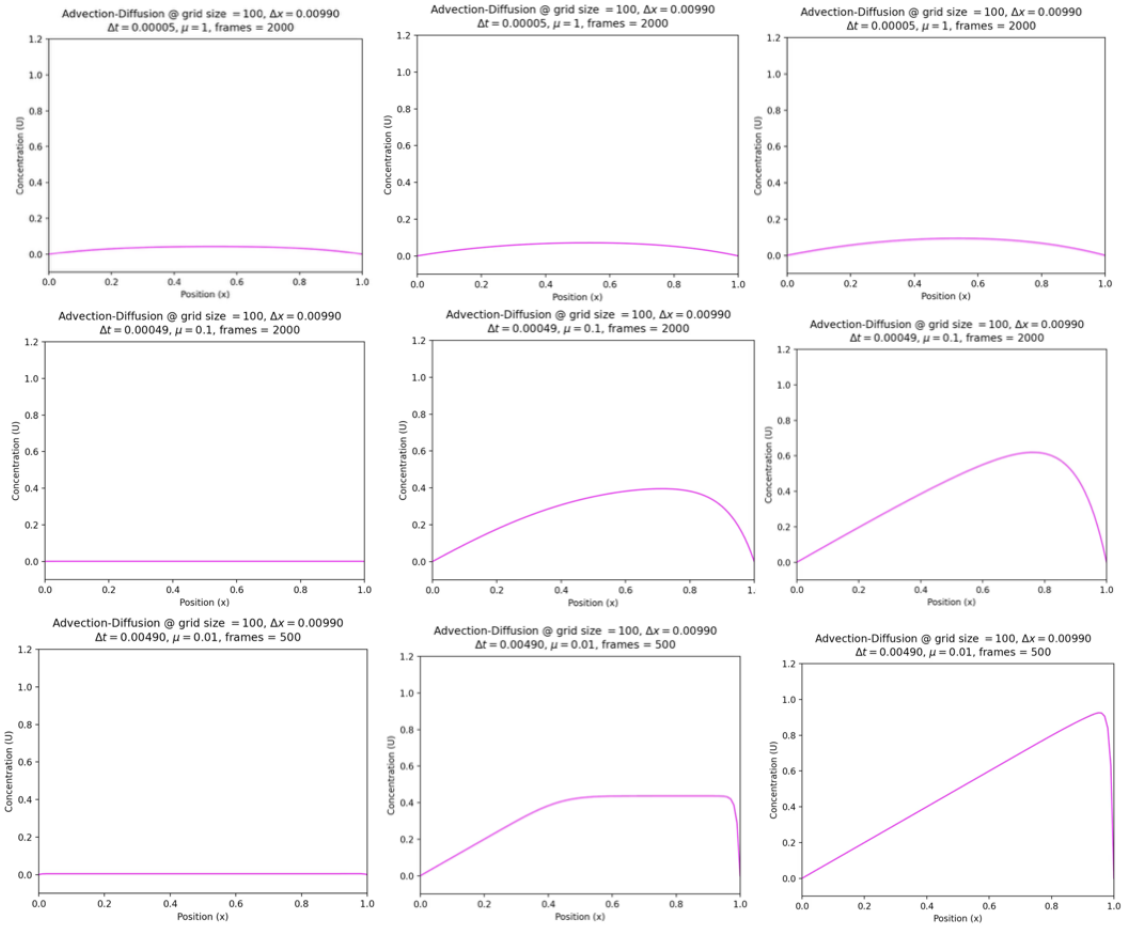


Figure 8: Advection-Diffusion simulations at grid size = 100, with $\Delta x = 0.0099, \Delta t = 0.00005$ with diffusion coefficients $\mu = 1$(top row),$\mu = 0.1$, $\mu = 0.01$(bottom row). **Left:** time frame at the beginning; **Middle:** time frame in the middle; **Right:** time frame at the end.

Proof. Those pictures seem very similar, compared to the explicit method (exercise 3). To see better about the difference in computational speed, we also have the comparison movies below:

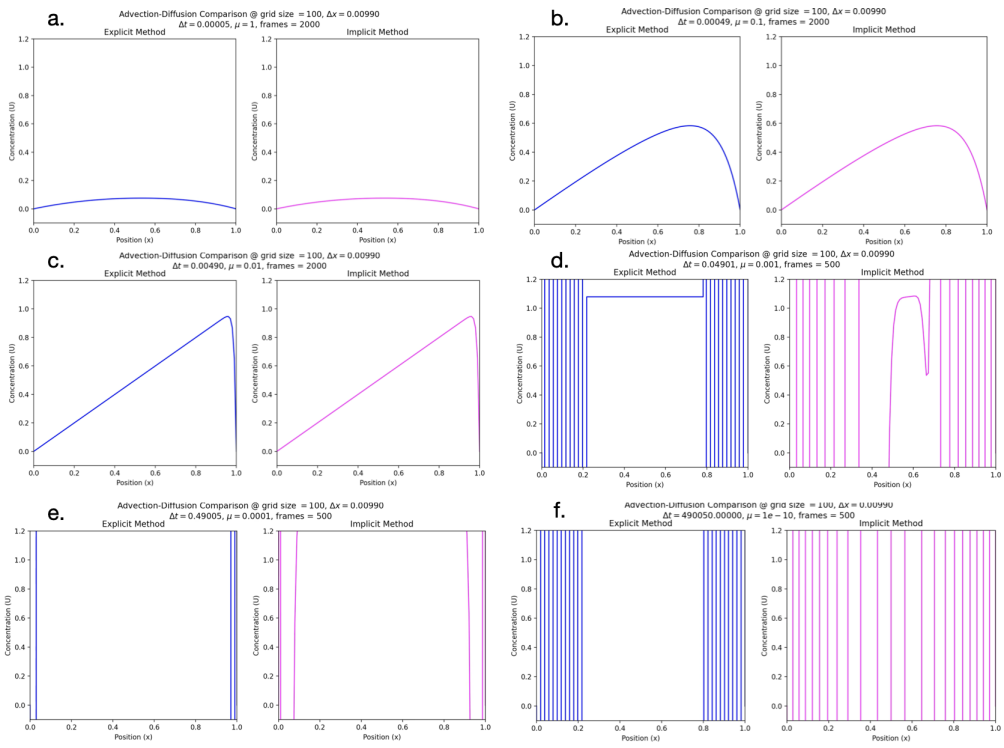


Figure 9: Advection-Diffusion process comparison (blue: explicit; magenta: implicit) **a:** $\mu = 1$; **b:** $\mu = 0.1$; **c:** $\mu = 0.01$; **d:** $\mu = 0.001$; **e:** $\mu = 0.0001$; **f:** $\mu = 1e - 10$

.

Now the speed and difference is more pronounced that at and after $\mu = 0.001$. Implicit method is definitely faster, compared to the explicit method. It also seems to be better about dealing with boundary layers and instability. This is because $\Delta(t) = \mathcal{O}(\Delta x)$ can give stability, whereas in exercise 2, we only have $\mathcal{O}(\Delta x^2)$. So when μ is smaller and Δt is bigger, we can get a better result with the implicite method. □

7 Exercise 6

7.1 Try to find time stepping scheme using D_0 and D_+D_- that is second order accurate and stable with an "advective" time step $\Delta t = \lambda \Delta x$. One possibility is a split scheme that does the advective part using an explicit four state Runge Kutta method and the diffusive part using Crank Nicholson. Does this work (is it stable and second order)?

Proof. We used a split scheme here with a 4 stage RK4 for the advenction process and the crank nicholson method for the diffusion process. In order to maintain the stability, we need to change the CFL condition (Jameson Method):

$$\mu \frac{\Delta t}{\Delta x^2} \leq \frac{1}{2} \Longrightarrow \frac{a\Delta t}{\Delta x} \leq 2\sqrt{2}, \quad a = 2.8 \tag{59}$$

Then, we have the below plots:

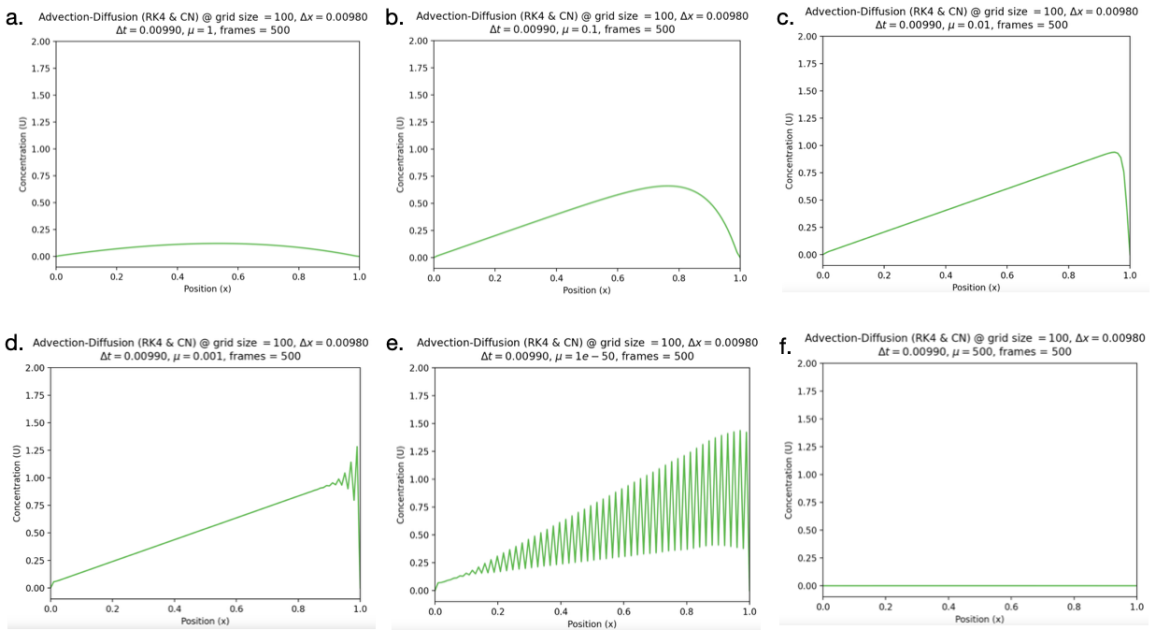


Figure 10: Advection-diffusion with $n = 100$. **a:** $\mu = 1$, **b:** $\mu = 0.1$, **c:** $\mu = 0.01$, **d:** $\mu = 0.001$, **e:** $\mu = 1e - 50$, **f:** $\mu = 500$

As we can see, even with very extreme cases $\mu = 1e - 50$, $\mu = 500$, our movies still remain stable, where μ is our Lyapunov exponent that measures the stability of the system. In order to give an explanation, we start with our PDE:

$$\partial_t u + a\partial_x u = \mu \partial_x^2 u + S(x) \Longleftrightarrow \partial_t u = [ivk - \mu k^2]\hat{u}(k,t) \Longleftrightarrow \partial_t u = [-v\partial_x + \mu \partial_x^2]u(\vec{x},t) \tag{60}$$

where $\hat{F}_s = [-v\partial_x + \mu\partial_x^2]$ and $\hat{F}_p = [ivk - \mu k^2]$ are our force operators in the spatial and phase domains. Along the way, now we understand that the forces acting on the system are composed of advection and diffusion components, where they compete with each other (they have different signs), in the spatial and phase domain. The same thing applies in the CFL condition:

$$CFL_{adv} = \frac{a\Delta x}{\Delta t}, \quad CFL_{diff} = \frac{\Delta t}{\Delta x^2} \Longleftrightarrow CFL = f(CFL_{adv}, CFL_{diff}) \Longleftrightarrow \Delta t = \min\left(\frac{\Delta x}{a}, \frac{\Delta x^2}{2\mu}\right) \tag{61}$$

We have tried using CFL_{diff} to optimize the system. However, it doesn't work, which tells us we should focus on CFL_{adv}. We start with our advection process:

$$\partial_t u + a\partial_x u = 0 \Longleftrightarrow \dot{U}_j = -aD_0U_j \Longleftrightarrow U_j(t) = c(t)e^{i\theta j} \Longleftrightarrow U_j(t) = e^{it\sin(\theta)}e^{i\theta j} \tag{62}$$

where $e^{i\theta j} = i/\Delta x \sin(\theta)$, and $|e^{it\sin(\theta)}| = 1$ means stable. Moreover, we also know that, in Forward Euler:

$$U_j^{(k+1)} = U_j^{(k)} - \Delta t a D_0 U_j^{(k)} \Longleftrightarrow U_j^{(k+1)} = [1 - \Delta t a D_0]U_j^{(k)} \Longleftrightarrow U_j^{(k+1)} = MU_j^{(k)} \tag{63}$$

Hence, we have:

$$m(\theta) = 1 - a\frac{\Delta t}{\Delta x} i\sin(\theta) \Longleftrightarrow |m(\theta)|^2 = 1 + \frac{a\Delta t}{\Delta x}\sin^2(\theta) > 1. \text{ unless } \theta = 0 \tag{64}$$

Using Jameson's method, we can get $a = 2.8$, with $a\Delta t/\Delta x \leq 2\sqrt{2}$, which satisfies the bound and shows the stability. Now, we want to show that the algorithm is second order accurate. As we know, RK4 has a local truncation error of $\mathcal{O}(\Delta t^5)$ with a gloabl error of $\mathcal{O}(\Delta x^4)$. Crank-Nicolson has both local and global truncation errors of $\mathcal{O}(\Delta t^2)$in time and $\mathcal{O}(\Delta x^2)$, which gives $\mathcal{O}(\Delta t^2 + \Delta x^2)$. Moreover, we know that for a splitting process, we have:

$$\dot{x} = f(\vec{x}) + g(\vec{x}) \Longleftrightarrow \phi_{f+g}(x, \Delta t) = \phi_f(x, \Delta t) \circ \phi_g(x, \Delta t) + \mathcal{O}(\Delta t^2) \tag{65}$$

Moreover, we also know that:

$$\begin{aligned} x(t + \Delta t) = e^{\Delta t A}x(t) &\Longleftrightarrow e^{\Delta t}(A + B) + \frac{\Delta t^2}{2}(A + B)^2 + \mathcal{O}(\Delta t^3) \\ &\Longrightarrow (I + \Delta t\text{CN})[(I - \Delta t\text{RK4})u^k] - (I - \Delta t\partial_x + \Delta t\partial_x^2)u^k \\ &= u^k - \Delta t(\partial_x)^k + O(\Delta t^6) + \Delta t[(\partial_x^2)^k + O(\Delta t^2)] - (I - \Delta t\partial_x + \Delta t\partial_x^2)u^k \\ &= u^k - \Delta t(\partial_x)^k + \Delta t(\partial_x^2)^k - (I - \Delta t\partial_x + \Delta t\partial_x^2)u^k + O(\Delta t^3) \end{aligned} \tag{66}$$

with A $= (I + \Delta t\text{CN})$ and $B = (I - \Delta t\text{RK4})$, and we showed that we have a second order accuracy. □